

Loanword Adaptation Algorithm for Filipino

Zhean Robby Ganituen
De La Salle University
Manila, Philippines

zhean_robbly_ganituen@dlsu.edu.ph

Clarence Ivan Ang
De La Salle University
Manila, Philippines

clarence_ang@dlsu.edu.ph

Erin Gabrielle Chua
De La Salle University
Manila, Philippines

erin_chua@dlsu.edu.ph

Justin Ching
De La Salle University
Manila, Philippines

justin_ethan_l_ching@dlsu.edu.ph

Jaztin Jacob Jimenez
De La Salle University
Manila, Philippines

jaztin_jacob_jimenez@dlsu.edu.ph

Roan Cedric Campo
De La Salle University
Manila, Philippines

roan_campo@dlsu.edu.ph

Clive Jarel Ang
De La Salle University
Manila, Philippines

clive_ang@dlsu.edu.ph

Nathaniel Oco
De La Salle University
Manila, Philippines

nathaniel.oco@dlsu.edu.ph

Abstract—This is the abstract.
Index Terms—one, two, three

I. INTRODUCTION

Borrowing words from one language (or the source language L_2) to another (the borrowing language L_1) is a common linguistic phenomenon. For English, the word *modus operandi* may be used in criminal investigations. In the context of shipping, one may use *cruise*, *dock*, or *yacht*, all of which are from Dutch.

Borrowing is more common in locales where multiple languages are being used simultaneously in similar contexts. An example of this is the Philippine context: which has both Filipino ($\mathcal{F}$) and English ($\mathcal{E}$) as their official language. In casual contexts, like going to school, one may use *paaralan* which is the native Filipino word for “school”. Or borrow by using *eskuwelahan* from Spanish, or *iskul* or *skul* from English. Furthermore, the Philippines also has a rich history of borrowings from Malay and Chinese way before they started borrowing from Spanish and English (during their colonial period) [1].

In general, L_1 borrows a word $w \in L_2$, we consider how the w is pronounced and written in L_2 and constraints in L_1 [2], [3]. Using the example above, we look at the base pronunciation and spelling of *school*, which are /sku:l/ and “school”. With the relevant Filipino constraints: “i/e-” prefix for English or Spanish loanwords and strict phoneme-grapheme mapping. Which is why the adapted loanword is *iskul*.

The algorithmic task of transforming $w \in L_2$ to its adapted form $\omega \in \mathcal{F}$ has importance in Filipino-centric NLP where loanword adaptation or code-switching occurs, like spell and grammar checking for Filipino text data with loanwords which is common in a most contexts such as social media posts, casual conversations, and technical settings where using the native words may be difficult. Or, text-to-speech or sentence generation for Taglish.

Finally, even though Filipino has a orthography manual publicly available, no software solution is yet to be implemented for writing adapted loanwords.

Hence, the goal of the project is to study loanword adaptation in Filipino and implementing an adaptation algorithm.

A. Phonology of Filipino

B. Loanwords in Filipino

II. PROBLEM DEFINITION

III. GENERATIVE PHONOLOGY

IV. PROPOSED ALGORITHM

V. EXPERIMENTAL SETUP

VI. RESULTS AND DISCUSSION

VII. CONCLUSION AND FUTURE WORK

REFERENCES

- [1] E. Baklanova, “Types of borrowings in tagalog/filipino,” *Kritika Kultura*, no. 28, Mar. 2017. [Online]. Available: <http://journals.ateneo.edu/ojs/index.php/kk/article/view/KK2017.02803/2407>
- [2] C. Uffman, *Loanword Adaptation*. Oxford University Press, Nov. 2015, p. 644–666. [Online]. Available: <https://doi.org/10.1093/oxfordhb/9780199232819.013.019>
- [3] L. Mao and M. Hulden, “How regular is Japanese loanword adaptation? a computational study,” in *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, Y. Matsumoto and R. Prasad, Eds. Osaka, Japan: The COLING 2016 Organizing Committee, Dec. 2016, pp. 847–856. [Online]. Available: <https://aclanthology.org/C16-1081/>